

P-EDCA under light load, Dual DS-CTS: fixes low-k timing, costs high-k airtime

802.11be, nSta=30, CWds×QSRC×PSRC sweep | offered load 0.1 / 0.5 / 1.0 Mbps per STA | P-EDCA STA P99

The three load regimes (EDCA-only VO P99 sets the scene)

■ 0.1 Mbps — uncongested

EDCA-only P99 = 0.99 ms (sub-ms). Little for P-EDCA to fix.

■ 0.5 Mbps — congested, not saturated

EDCA-only P99 = 5.80 ms. Real queueing; whether P-EDCA helps now depends on k.

■ 1.0 Mbps — saturated

EDCA-only P99 = 10.88 ms. Far lower than the pre-fix baseline (NAV/CF-End bugs fixed).

0.1 Mbps: +1~+20%

modest but consistently positive at every k

n=15 weakest (+1%) — winning combo still drifts, params matter less here

0.5 & 1.0 Mbps: sign flips at n=30

n=5 +37%/+29% → n=30 -3.5%/-1.4%

full penetration: P-EDCA now COSTS more than it saves

Recipe must now scale with k

QSRC rises, PSRC falls as k grows (opposite of pre-Dual)

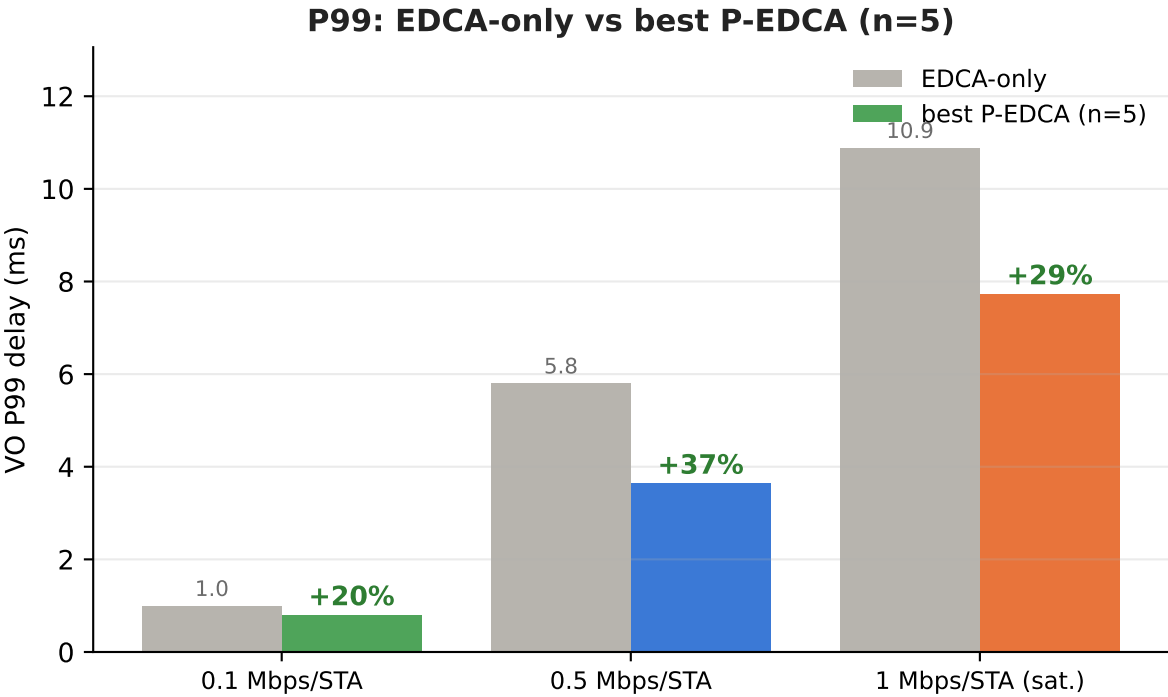
one fixed combo can't cover k=5..30 any more

One-line takeaway

Dual DS-CTS fixed the old low-k timing-miss failure, so P-EDCA now helps at k=5-15 everywhere. But the DS-CTS reservation is exclusive (one window, one transmitter), so cost scales with k while benefit doesn't -- at k=30 it now costs more than it saves.

Best P-EDCA P99 across loads — gain peaks at low-moderate penetration

EDCA-only baseline vs best-of-36-combos P-EDCA (P-EDCA STAs, n=5), with % P99 change



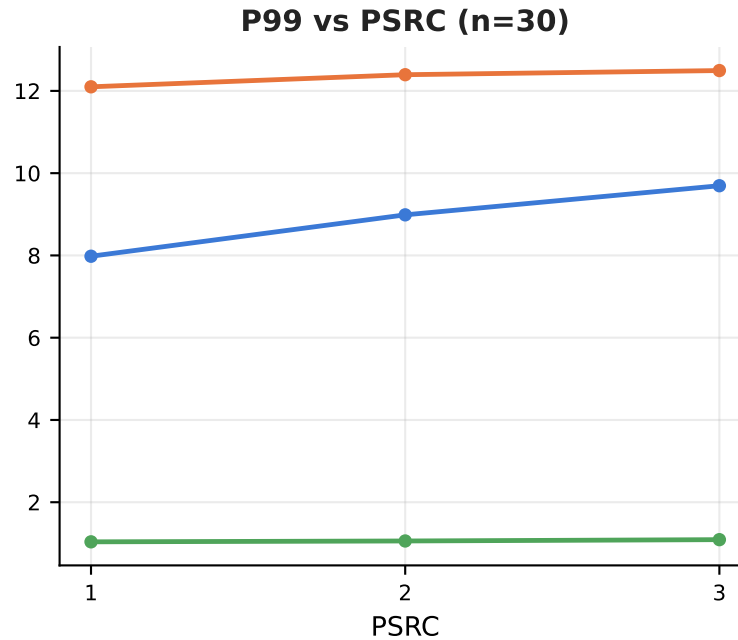
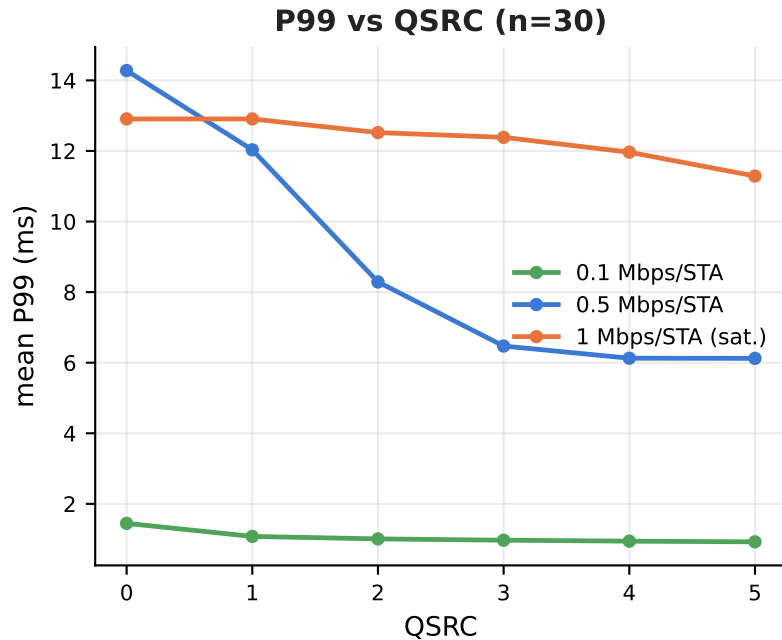
Best combo & P99 per P-EDCA population

Load	nPedca	best c/q/s	P99 (ms)
0.1 Mbps/STA	5	c0 q3 s1	0.79 (+20%)
	15	c1 q3 s2	0.98 (+1%)
	30	c0 q5 s1	0.85 (+14%)
0.5 Mbps/STA	5	c1 q0 s3	3.63 (+37%)
	15	c1 q2 s2	5.42 (+7%)
	30	c0 q5 s1	6.00 (-4%)
1 Mbps/STA (sat.)	5	c1 q1 s3	7.72 (+29%)
	15	c1 q5 s3	8.85 (+19%)
	30	c0 q5 s1	11.03 (-1%)

- Gain no longer rises monotonically with penetration -- it PEAKS at low-moderate k and reverses at full penetration: 0.5 Mbps +37% (n=5) → +7% (n=15) → -3.5% (n=30); 1.0 Mbps +29% → +19% → -1.4%.
- Why: the DS-CTS reservation is exclusive -- one protected window serves exactly one transmitter, but every P-EDCA STA pays the DS-CTS airtime cost to compete for it. Benefit per STA scales ~1/k while cost scales ~k, so net value crosses zero as k grows.
- At 0.1 Mbps gain stays small and positive at every k (+1~+20%) -- uncongested traffic never reaches the exclusivity bottleneck, so the reservation cost stays affordable.

Which knob matters at which k — QSRC and PSRC reversed direction

Marginal mean P99 (P-EDCA STAs, n=30) vs QSRC and vs PSRC, per load



k-driven recipe

CWds = 1 fixed; QSRC/PSRC scale with k

k=5 -> QSRC=2, PSRC=3

k=15 -> QSRC=4, PSRC=2

k=30 -> QSRC=5, PSRC=1

law: $\text{QSRC} = \text{round}(0.5 + 0.2k)$, $\text{PSRC} = 3/2/1$
for $k \leq 10$ / ≤ 22 / > 22

- QSRC direction REVERSED from the pre-Dual deck: at n=30, LOW QSRC is now worst (q0 14.3 ms) and HIGH QSRC is best (q5 6.1 ms) at 0.5 Mbps -- q0 keeps every STA retrying into the exclusive window; higher QSRC filters out most of the excess demand before it pays the DS-CTS cost.
- PSRC direction also reversed at high k: s1 8.0 ms now beats s3 9.7 ms at n=30 -- fewer consecutive attempts per winner leaves more of the shared airtime for everyone else. At n=5 the OLD direction still holds (PSRC=3 better) since exclusivity barely bites when only 5 STAs compete.
- Recommendation: stop using one fixed combo. Estimate k (number of STAs actually using P-EDCA) and apply the k-driven law above; getting k badly wrong costs more than getting traffic type wrong (see memory "pedca-adaptive-policy" for the full cost table and the closed-loop controller that automates this).